

More Logarithmic Equations and Properties of Logarithms

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$
or $\sqrt[a]{x^b}$ or $\sqrt[7]{x^5}$

More Complicated Logarithmic Equations

Solve. Write the Roman numeral of each exponent rule that you use.

① $3 \log_2 x - 8 = 1$

$3 \log_2 x = 9$

$\log_2 x = 3$

$2^3 = x$

$\boxed{8 = x}$

③ $4 = 7 \log_x 9 + 3$

$1 = 7 \log_x 9$

$\frac{1}{7} = \log_x 9$

$x^{\frac{1}{7}} = 9$

$\sqrt[7]{x} = 9$

$(\sqrt[7]{x})^7 = (9)^7$

$\boxed{x = 9^7}$

IX

② $6 = \frac{\log_x 5}{2} + 1$

$5 = \frac{\log_x 5}{2}$

$10 = \log_x 5$

$x^{10} = 5$

$\sqrt[10]{x^{10}} = \pm \sqrt[10]{5}$

$x = \pm \sqrt[10]{5}$

$\boxed{x = \sqrt[10]{5}}$

④ $-6 - \frac{\log_4 x}{3} = -5$

$-\frac{\log_4 x}{3} = 1$

$\log_4 x = -3$

$4^{-3} = x$

$\frac{1}{4^3} = x$

$\boxed{\frac{1}{64} = x}$

VIII

VI

$$\textcircled{5} 10 + 4\log_6 X = 14$$

$$4\log_6 X = 4$$

$$\log_6 X = 1$$

$$6^1 = X$$

$$\boxed{6 = X}$$

IX

$$\textcircled{7} 17 = 32\log_x 8 + 1$$

$$16 = 32\log_x 8$$

$$\frac{1}{2} = \log_x 8$$

$$x^{\frac{1}{2}} = 8$$

$$\sqrt{x} = 8$$

$$(\sqrt{x})^2 = (8)^2$$

$$\boxed{x = 64}$$

VIII

$$\textcircled{6} \frac{\log_x 81}{4} - 11 = -10$$

$$\frac{\log_x 81}{4} = 1$$

$$\log_x 81 = 4$$

$$x^4 = 81$$

$$\sqrt[4]{x^4} = \pm \sqrt[4]{81}$$

$$x = \pm 3$$

$$\boxed{x = 3}$$

$$\textcircled{8} \frac{\log_7 X}{6} + 5 = 5$$

$$\frac{\log_7 X}{6} = 0$$

$$\log_7 X = 0$$

$$7^0 = X$$

$$\boxed{1 = X}$$

More Properties of Logarithms

Logarithm Property V

$$\log_k(AB) = \log_k A + \log_k B$$

Logarithm Property VI

$$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$$

Convert to a sum or difference of logarithms.

$$\textcircled{9} \log_2(6X) = \boxed{\log_2 6 + \log_2 X}$$

$$\textcircled{10} \log_5\left(\frac{Y}{3}\right) = \boxed{\log_5 Y - \log_5 3}$$

$$\textcircled{11} \log_4(XY) = \boxed{\log_4 X + \log_4 Y}$$

$$\textcircled{12} \log_7\left(\frac{8}{b}\right) = \boxed{\log_7 8 - \log_7 b}$$

$$\textcircled{13} \log_3(4Y) = \boxed{\log_3 4 + \log_3 Y}$$

$$\textcircled{14} \log_{12}\left(\frac{6}{X}\right) = \boxed{\log_{12} 6 - \log_{12} X}$$

$$\textcircled{15} \log_8(mn) = \boxed{\log_8 m + \log_8 n}$$

$$\textcircled{16} \log_5\left(\frac{L}{P}\right) = \boxed{\log_5 L - \log_5 P}$$

Convert each sum or difference to one logarithm.

$$(17) \log_4 7 + \log_4 3 = \log_4 (7 \cdot 3) = \boxed{\log_4 21}$$

$$(18) \log_2 X - \log_2 5 = \boxed{\log_2 \left(\frac{X}{5}\right)}$$

$$(19) \log_3 M + \log_3 C = \boxed{\log_3 (MC)}$$

$$\star (20) -\log_9 2 + \log_9 8 = \log_9 8 - \log_9 2 = \log_9 \left(\frac{8}{2}\right) = \boxed{\log_9 4}$$

$$(21) \log_6 a + \log_6 b = \boxed{\log_6 (ab)}$$

$$(22) \log_5 3 - \log_5 6 = \log_5 \left(\frac{3}{6}\right) = \boxed{\log_5 \left(\frac{1}{2}\right)}$$

$$(23) \log_8 9 + \log_8 10 = \log_8 (9 \cdot 10) = \boxed{\log_8 90}$$

$$\star (24) -\log_{15} N + \log_{15} 10 = \log_{15} 10 - \log_{15} N = \boxed{\log_{15} \left(\frac{10}{N}\right)}$$

Logarithm Property VII

$$\log_k A^B = B \log_k A$$

Convert each logarithm to a multiple of a logarithm.

$$(25) \log_3 5^7 = \boxed{7 \log_3 5}$$

$$(26) \log_8 9^4 = \boxed{4 \log_8 9}$$

$$(27) \log_x 17^3 = \boxed{3 \log_x 17}$$

$$(28) \log_9 Y^{10} = \boxed{10 \log_9 Y}$$

$$(29) \log_4 8^5 = \boxed{5 \log_4 8}$$

$$(30) \log_5 7^9 = \boxed{9 \log_5 7}$$

$$(31) \log_6 X^8 = \boxed{8 \log_6 X}$$

$$(32) \log_y 3^{16} = \boxed{16 \log_y 3}$$

Convert each expression from a multiple of a logarithm to a logarithm of a power.

$$(33) 8\log_3 X = \boxed{\log_3 X^8}$$

$$(34) 2\log_5 8 = \log_5 8^2 = \boxed{\log_5 64}$$

$$(35) 7\log_4 6 = \boxed{\log_4 6^7}$$

$$(36) 5\log_6 Y = \boxed{\log_6 Y^5}$$

$$(37) 3\log_9 N = \boxed{\log_9 N^3}$$

$$(38) 3\log_2 3 = \log_2 3^3 = \boxed{\log_2 27}$$

$$(39) 6\log_5 X = \boxed{\log_5 X^6}$$

$$(40) 8\log_3 8 = \boxed{\log_3 8^8}$$

Using Properties V-VII

Convert each logarithm to a sum or difference of multiples of logarithms. Write the Roman numeral of each logarithm property that you use.

$$(41) \log_4(3x^2)$$

V

VII

$$\log_4 3 + \log_4 x^2$$

$$\boxed{\log_4 3 + 2\log_4 X}$$

VI

VII

$$(43) \log_3(bc^3)$$

V

VII

$$\log_3 b^2 + \log_3 c^3$$

$$\boxed{2\log_3 b + 3\log_3 c}$$

VI

VII

$$(45) \log_6(9m^5)$$

V

VII

$$\log_6 9 + \log_6 m^5$$

$$\boxed{\log_6 9 + 5\log_6 M}$$

VI

VII

$$(47) \log_2(x^6y^5)$$

V

VII

$$\log_2 x^6 + \log_2 y^5$$

$$\boxed{6\log_2 X + 5\log_2 Y}$$

VI

VII

$$(42) \log_5\left(\frac{Y^6}{4}\right)$$

$$\log_5 Y^6 - \log_5 4$$

$$\boxed{6\log_5 Y - \log_5 4}$$

$$(44) \log_8\left(\frac{a^7}{b^4}\right)$$

$$\log_8 a^7 - \log_8 b^4$$

$$\boxed{7\log_8 a - 4\log_8 b}$$

$$(46) \log_7\left(\frac{2}{y^8}\right)$$

$$\log_7 2 - \log_7 y^8$$

$$\boxed{\log_7 2 - 8\log_7 Y}$$

$$(48) \log_3\left(\frac{a^2}{b^5}\right)$$

$$\log_3 a^2 - \log_3 b^5$$

$$\boxed{2\log_3 a - 5\log_3 b}$$

$$\textcircled{49} \log_4(6x^7y^8)$$

$$\text{V} \log_4(6x^7) + \log_4 y^8$$

$$\log_4(6x^7) + \log_4 y^8$$

$$\text{V} \log_4 6 + \log_4 x^7 + \log_4 y^8$$

$$\text{VII} \boxed{\log_4 6 + 7\log_4 x + 8\log_4 y}$$

$$\textcircled{50} \log_9(4x^2y^{10})$$

$$\text{V} \log_9(4x^2) + \log_9 y^{10}$$

$$\log_9(4x^2) + \log_9 y^{10}$$

$$\text{V} \log_9 4 + \log_9 x^2 + \log_9 y^{10}$$

$$\text{VII} \boxed{\log_9 4 + 2\log_9 x + 10\log_9 y}$$

One of the Most Common Mistakes!

Not Okay

$$\log_4(3x^2)$$

$$2\log_4(3x)$$

Okay

$$\log_4(3x)^2$$

$$2\log_4(3x)$$

$$\textcircled{51} \log_2\left(\frac{a^5}{3b^4}\right)$$

$$\text{VI} \log_2 a^5 - \log_2(3b^4)$$

$$\log_2 a^5 - \log_2(3b^4)$$

$$\text{V} \log_2 a^5 - (\log_2 3 + \log_2 b^4)$$

$$\log_2 a^5 - \log_2 3 - \log_2 b^4$$

$$\text{VII} \boxed{5\log_2 a - \log_2 3 - 4\log_2 b}$$

$$\textcircled{52} \log_4\left(\frac{7b^5}{c^3}\right)$$

$$\text{VI} \log_4(7b^5) - \log_4 c^3$$

$$\log_4(7b^5) - \log_4 c^3$$

$$\text{V} \log_4 7 + \log_4 b^5 - \log_4 c^3$$

$$\text{VII} \boxed{\log_4 7 + 5\log_4 b - 3\log_4 c}$$

Convert each expression to one logarithm.

$$\textcircled{53} \log_3 x + 2\log_3 y$$

$$\text{VII} \log_3 x + \log_3 y^2$$

$$\text{V} \boxed{\log_3(xy^2)}$$

$$\textcircled{54} 6\log_5 m - \log_5 n$$

$$\log_5 m^6 - \log_5 n$$

$$\text{VI} \boxed{\log_5\left(\frac{m^6}{n}\right)}$$

$$\textcircled{55} \quad 3\log_2 b + 9\log_2 c \quad \star \textcircled{56} \quad -6\log_8 X + 5\log_8 Y$$

$$\text{VII} \quad \log_2 b^3 + \log_2 c^9 \quad 5\log_8 Y - 6\log_8 X$$

$$\text{V} \quad \boxed{\log_2(b^3 c^9)} \quad \text{VII} \quad \log_8 Y^5 - \log_8 X^6$$

$$\text{VI} \quad \boxed{\log_8 \left(\frac{Y^5}{X^6} \right)}$$

$$\textcircled{57} \quad 4\log_5 b + \log_5 c \quad \textcircled{58} \quad \log_3 M - 9\log_3 L$$

$$\text{VII} \quad \log_5 b^4 + \log_5 c \quad \text{VII} \quad \log_3 M - \log_3 L^9$$

$$\text{V} \quad \boxed{\log_5(b^4 c)} \quad \text{VI} \quad \boxed{\log_3 \left(\frac{M}{L^9} \right)}$$

$$\textcircled{59} \quad 6\log_7 b + 5\log_7 c \quad \star \textcircled{60} \quad -3\log_7 X + 8\log_7 Y$$

$$\text{VII} \quad \log_7 b^6 + \log_7 c^5 \quad 8\log_7 Y - 3\log_7 X$$

$$\text{V} \quad \boxed{\log_7(b^6 c^5)} \quad \text{VII} \quad \log_7 Y^8 - \log_7 X^3$$

$$\text{VI} \quad \boxed{\log_7 \left(\frac{Y^8}{X^3} \right)}$$

$$\textcircled{61} \quad \log_5 c + 3\log_5 b + 2\log_5 a$$

$$\text{VII} \quad \log_5 c + \log_5 b^3 + \log_5 a^2$$

$$\text{V} \quad \log_5(c b^3) + \log_5 a^2$$

$$\log_5(c b^3) + \log_5 a^2$$

$$\text{V} \quad \boxed{\log_5(c b^3 a^2)}$$

$$\textcircled{62} \quad 6\log_8 X + \log_8 c + 7\log_8 n$$

$$\text{VII} \quad \log_8 X^6 + \log_8 c + \log_8 n^7$$

$$\text{V} \quad \log_8(X^6 c) + \log_8 n^7$$

$$\log_8(X^6 c) + \log_8 n^7$$

$$\text{V} \quad \boxed{\log_8(X^6 c n^7)}$$

$$\textcircled{63} \quad 5\log_3 X - 2\log_3 Y + 8\log_3 M$$

$$\text{VII} \quad \log_3 X^5 - \log_3 Y^2 + \log_3 M^8$$

$$\text{VI} \quad \log_3 \left(\frac{X^5}{Y^2} \right) + \log_3 M^8$$

$$\log_3 \left(\frac{X^5}{Y^2} \right) + \log_3 M^8$$

$$\text{V} \quad \log_3 \left(\frac{X^5}{Y^2} \cdot M^8 \right) = \boxed{\log_3 \left(\frac{X^5 M^8}{Y^2} \right)}$$

$$\textcircled{64} \quad 3\log_9 X + 6\log_9 Y - \log_9 r$$

$$\text{VII} \quad \log_9 X^3 + \log_9 Y^6 - \log_9 r$$

$$\text{V} \quad \log_9 (X^3 Y^6) - \log_9 r$$

$$\log_9 (X^3 Y^6) - \log_9 r$$

$$\text{VI} \quad \boxed{\log_9 \left(\frac{X^3 Y^6}{r} \right)}$$

If $\log_k 2 \approx 0.4$ and $\log_k 3 \approx 0.6$, use this information along with Logarithm Properties V–VII to approximate the following values without a calculator. Write the Roman numeral of each logarithm property or exponent rule that you use.

$$\textcircled{65} \quad \log_k 6$$

$$\log_k (2 \cdot 3)$$

IV

$$\log_k 2 + \log_k 3$$

$$0.4 + 0.6$$

$$\boxed{1}$$

$$\textcircled{66} \quad \log_k \left(\frac{3}{2} \right)$$

VII

$$\log_k 3 - \log_k 2$$

$$0.6 - 0.4$$

$$\boxed{0.2}$$

$$\textcircled{67} \quad \log_k 8$$

$$\log_k (2^3)$$

VII

$$3\log_k 2$$

$$3(0.4)$$

$$\boxed{1.2}$$

$$\textcircled{68} \quad \log_k \sqrt{3}$$

E IX

$$\log_k (3^{\frac{1}{2}})$$

L VII

$$\frac{1}{2} \log_k 3$$

$$\frac{1}{2} (0.6)$$

$$\boxed{0.3}$$

$$\textcircled{69} \log_k 12$$

$$\log_k (4 \cdot 3)$$

LVI

$$\text{LVI} \log_k 4 + \log_k 3$$

$$\log_k (2 \cdot 2) + \log_k 3$$

$$\text{LVI} \log_k 2 + \log_k 2 + \log_k 3$$

$$0.4 + 0.4 + 0.6$$

$$\boxed{1.4}$$

$$\textcircled{71} \log_k 9$$

$$\log_k 3^2$$

EVII

$$\text{LVI} 2 \log_k 3$$

$$2(0.6)$$

$$\boxed{1.2}$$

$$\textcircled{72} \log_k \sqrt[3]{2}$$

$$\log_k (2^{\frac{1}{3}})$$

LVI

$$\frac{1}{3} \log_k 2$$

$$\frac{1}{3}(0.4)$$

$$\frac{1}{3}\left(\frac{2}{5}\right)$$

$$\boxed{\frac{2}{15}}$$

$$\textcircled{70} \log_k \left(\frac{2}{3}\right)$$

$$\log_k 2 - \log_k 3$$

$$0.4 - 0.6$$

$$\boxed{-0.2}$$

$$\textcircled{73} \log_k 18$$

$$\log_k (9 \cdot 2)$$

$$\text{LVI} \log_k 9 + \log_k 2$$

$$\log_k (3^2) + \log_k 2$$

EVII

$$\text{LVI} 2 \log_k 3 + \log_k 2$$

$$2(0.6) + 0.4$$

$$1.2 + 0.4$$

$$\boxed{1.6}$$

$$\textcircled{74} \log_k \left(\frac{1}{27}\right)$$

$$\log_k \left(\frac{1}{3^3}\right)$$

$$\log_k (3^{-3})$$

$$\text{LVI} -3 \log_k 3$$

$$-3(0.6)$$

$$\boxed{-1.8}$$